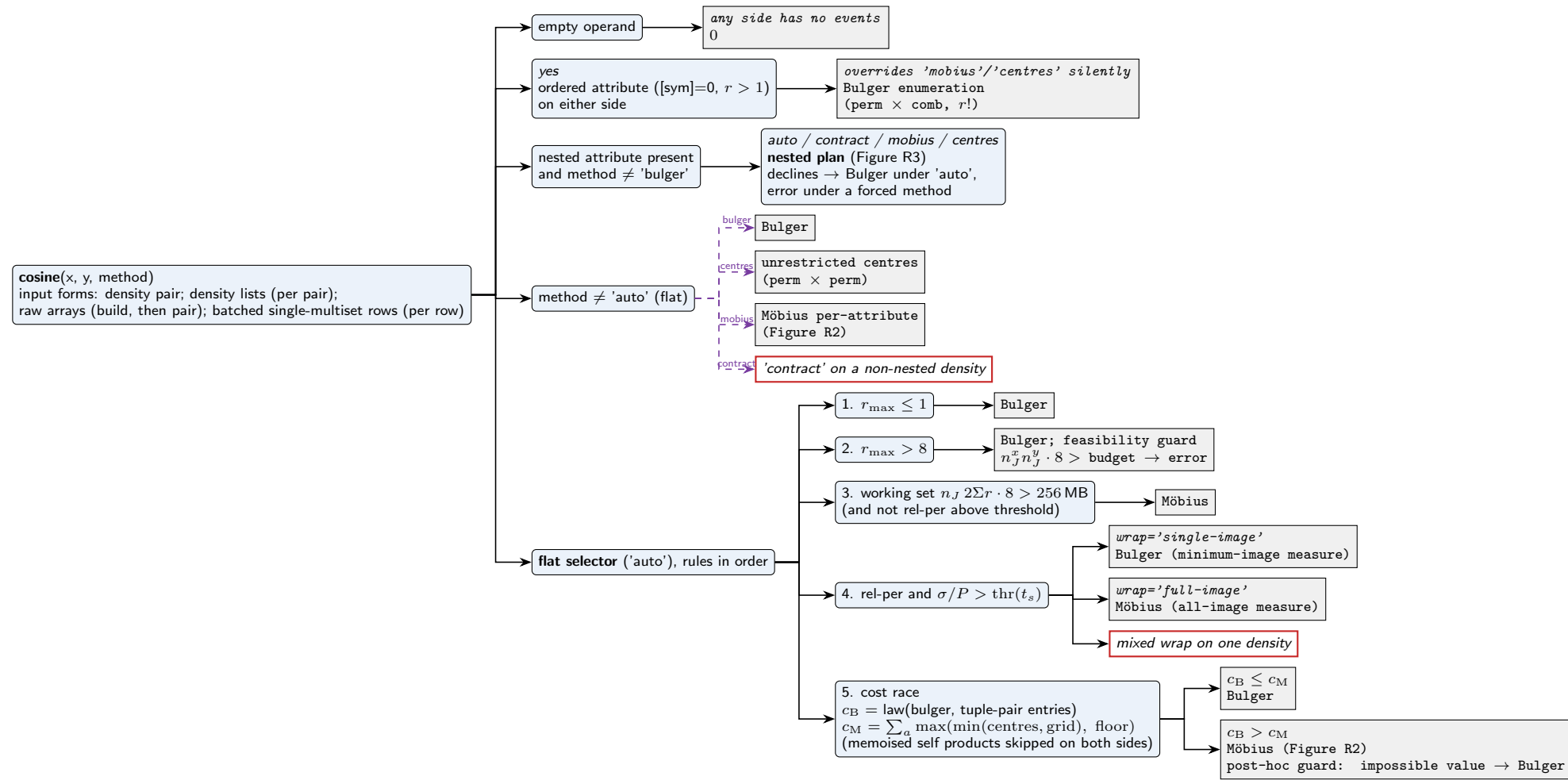
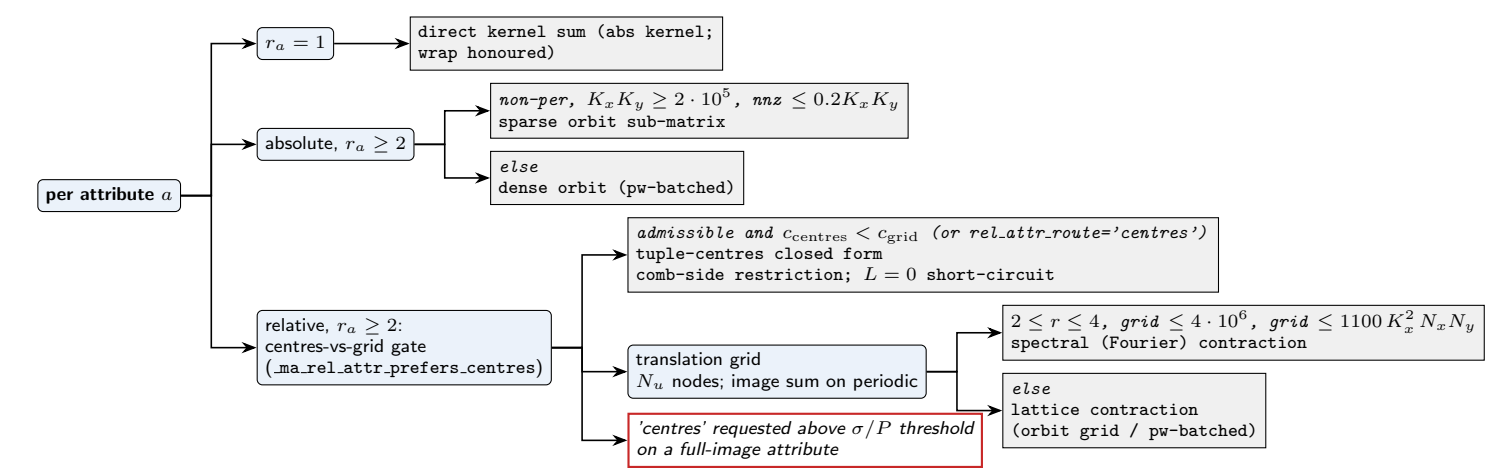


**Figure R1. Cosine similarity:  $\text{cos\_sim\_exp\_tens}$  /  $\text{cosSimExpTens}$ .** Rules are tested top to bottom; the first that fires decides. Dashed edges are **method overrides**. Both languages share every rule; only fitted constants differ.



**Kernel core shared by the Bulger and centres arms** (per attribute, log kernel over tuple pairs): Gram form when non-periodic and accurate enough; inner-[rel] block form when  $r_{\text{in}} > 0$ ; abs-per full-image  $\theta$  (wrapped Gaussian, image sum or Fourier) when wrap='full-image' and  $L > 0$ ; otherwise nearest-image reduction and the quadratic form  $Q$  (rel-per: pairwise minimum-image wrap). All- $r = 1$  densities take the direct kernel sum.

Figure R2. The Möbius arm: one  $(N_x \times N_y)$  matrix per attribute.



The centres route is blocked by measure when  $rel-per$  and  $\sigma/P > thr(t_s)$  (it computes the minimum-image reading), and by cost otherwise; a forced 'mobius' keeps the grid. Both routes memoise the self products under the key `mobius| $t_s$ |choices`.

Figure R3. The nested plan (`.try_nested_contract / nestedContract`).

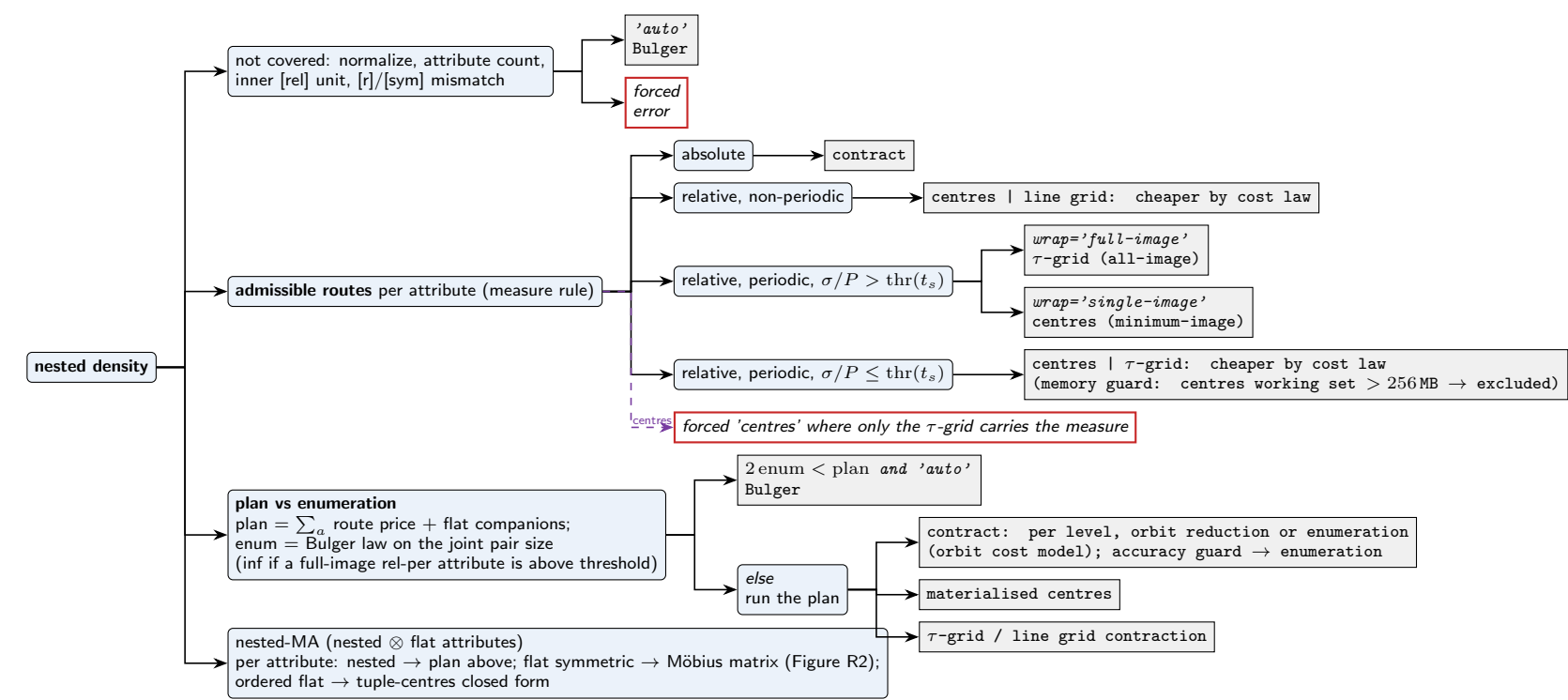
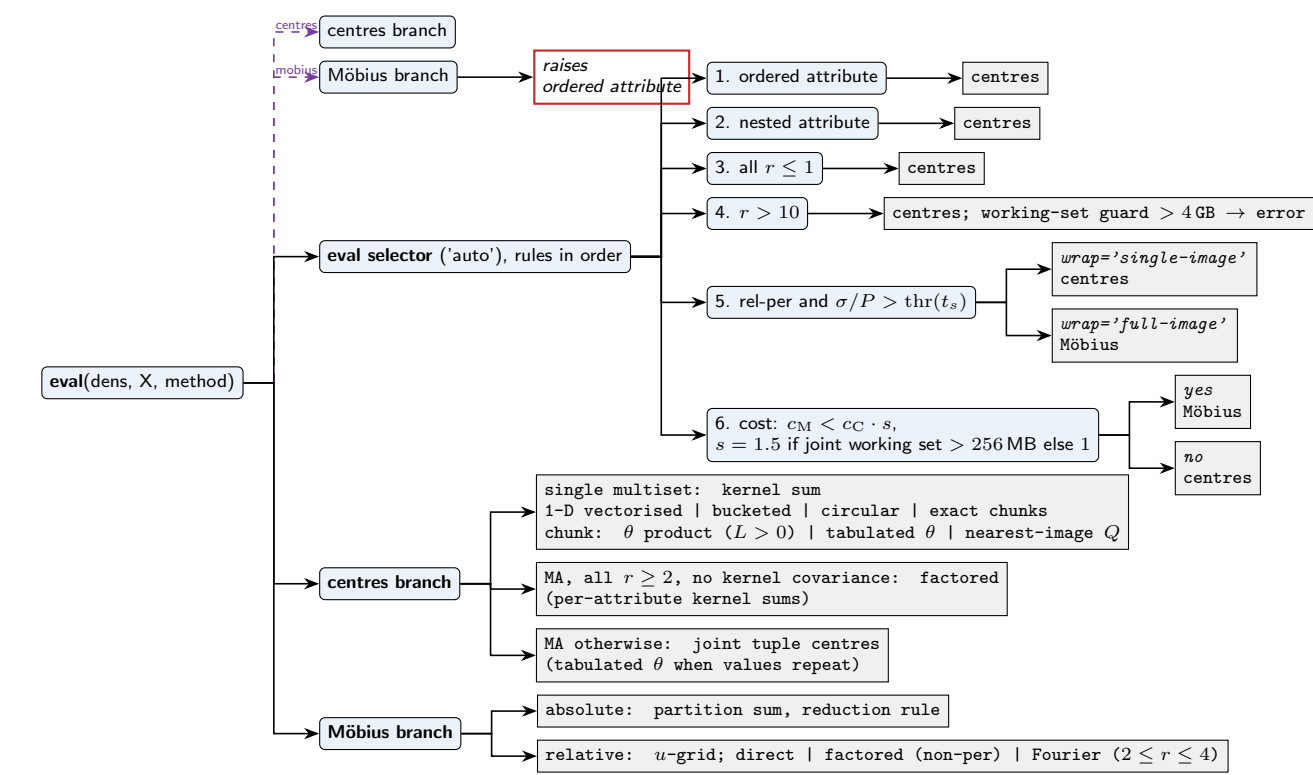


Figure R4. Point evaluation: eval\_exp\_tens / evalExpTens.



**Entropy** (`entropy_exp_tens`): 'differential' → adaptive grid of point evaluations (eval 'auto'); 'shannon' / 'normalized' → erf cell masses on absolute densities (minimum-image on periodic axes), grid via eval otherwise; 'renyi2' →  $\langle T, T \rangle$  from the per-attribute Möbius matrices (Figure R2), numerical for nested or ordered attributes. **Windowed similarity**: every window position is a cosine call under 'auto'; `windowed_tensor_similarity` has a single closed-form path. **Sweep**: mixture (dedup pairs) vs orbit sweep by pair count and chunk budget; the orbit sweep reuses Figure R2's per-attribute matrices.